Quantum Scientific Computing (QSC2026)

Homework 1

Assigned: Due: _____

Instructions. Ten problems on the fundamental concepts and methods of Lectures 1, 3, 4, and 5 (foundations; query algorithms and Grover search; the quantum Fourier transform and phase estimation; ground-state energy and amplitude estimation). Each is worth 10 points. Show your work: a correct number with no derivation earns little credit, and a clear argument with a small arithmetic slip earns most of it. You may use any result stated in the lecture notes, citing it by name. Collaboration is allowed on ideas, but write up your solutions independently.

Part I — Foundations (Lecture 1)

Problem 1. A single qubit: amplitudes, Bloch vector, and measurement. *(10 pts)* Let $|\psi\rangle = \cos\frac{\pi}{8}\,|0\rangle + e^{i\pi/4}\sin\frac{\pi}{8}\,|1\rangle$.

(a) Check that $|\psi\rangle$ is normalized and give the probabilities of outcomes 0 and 1 in a computational-basis measurement.

(b) Compute the expectation values $\langle\psi|X|\psi\rangle$, $\langle\psi|Y|\psi\rangle$, $\langle\psi|Z|\psi\rangle$ and hence the Bloch vector; verify it has unit length.

(c) Find the probability of outcome "+" when $|\psi\rangle$ is measured in the $\{|+\rangle, |-\rangle\}$ basis, and relate your answer to $\langle\psi|X|\psi\rangle$.

(d) *Sampling cost.* In practice $\langle Z\rangle$ is not read off exactly but *estimated*: each computational-basis measurement returns the eigenvalue $+1$ (outcome 0) or -1 (outcome 1), and one averages the results of N independent repetitions, $\hat{Z} = \frac{1}{N}\sum_{i=1}^{N} m_i$. The single-shot outcome $m_i \in \{\pm 1\}$ is random with variance $1 - \langle Z\rangle^2$, so $\hat{Z}$ fluctuates about $\langle Z\rangle$ with a standard error that shrinks only as $1/\sqrt{N}$ (the "shot-noise" or standard quantum limit). Using this scaling, estimate how many repetitions N are needed to determine $\langle Z\rangle$ to additive error $\epsilon = 0.05$ at about 95% confidence. State the scaling law (how N depends on ϵ) and show how your number follows from it.

Problem 2. Tensor products, entanglement, and the separability test. *(10 pts)* For each two-qubit state, use the separability criterion of §1.2.2 (a state $a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$ is a product state iff $ad = bc$) to decide whether it is a product state or entangled. If it is a product state, write it as $|u\rangle \otimes |v\rangle$; if entangled, say so.

(a) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$;

(b) $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$;

(c) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$.

(d) Prove directly that the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ admits no factorization $|u\rangle \otimes |v\rangle$.

Problem 3. Gates, circuits, reversibility, and no-cloning. *(10 pts)*

(a) Compute CNOT $(H \otimes \mathbb{I})\,|00\rangle$ and identify which Bell state it is.
(b) Show the Toffoli gate acts as $|a, b, 0\rangle \mapsto |a, b,\ a \wedge b\rangle$, a reversible AND. Why does reversibility require writing the result onto a fresh (ancilla) qubit rather than overwriting an input?
(c) Exhibit a two-qubit unitary U and a blank state $|s\rangle$ with $U\,|0\rangle\,|s\rangle = |0\rangle\,|0\rangle$ and $U\,|1\rangle\,|s\rangle = |1\rangle\,|1\rangle$, and explain why U fails to copy a general unknown qubit $\alpha\,|0\rangle + \beta\,|1\rangle$—i.e. why this is consistent with the no-cloning theorem.

Problem 4. Quantum teleportation, traced amplitude by amplitude. *(10 pts)* Alice wishes to teleport $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ to Bob; they share $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

(a) Write the three-qubit state before Alice's Bell measurement, expanded over her two-qubit measurement basis, and read off Bob's conditional (pre-correction) state for each of the four outcomes (m_1, m_2).
(b) For the outcome $(m_1, m_2) = (1, 1)$, state the correction gate Bob applies and verify it restores $|\psi\rangle$.
(c) Explain why the two classical bits are indispensable—that is, why Bob's qubit carries no information about $|\psi\rangle$ until the bits arrive (the no-signaling point).

Part II — Query algorithms and Grover search (Lecture 3)

Problem 5. Phase kickback, Deutsch, and Deutsch–Jozsa. *(10 pts)*

(a) For a Boolean f, show the phase-oracle identity $U_f\,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,|-\rangle$, where $U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.
(b) List the four functions $f : \{0,1\} \to \{0,1\}$, classify each as constant or balanced, and compute by hand the first-qubit measurement outcome produced by Deutsch's algorithm for each. Confirm it separates constant from balanced in a *single* query.
(c) In Deutsch–Jozsa on n bits, what is the probability of measuring the all-zeros outcome 0^n when f is constant? when f is balanced? State the promise that makes the one-query decision exact.

Problem 6. Grover search: reflections, geometry, and iteration count. *(10 pts)* Take $N = 16$ with a single marked item x_0, uniform start $|\psi_0\rangle = H^{\otimes 4}\,|0^4\rangle$.

(a) Writing $\sin(\theta/2) = 1/\sqrt{N}$, give θ and explain, using the two reflections $R_{x_0} = \mathbb{I} - 2\,|x_0\rangle\langle x_0|$ and $R_{\psi_0} = 2\,|\psi_0\rangle\langle\psi_0| - \mathbb{I}$, why one Grover iterate $G = R_{\psi_0}R_{x_0}$ rotates the state by θ in the plane $\mathrm{span}\{|x_0\rangle, |\psi_0\rangle\}$.
(b) The success probability after k iterates is $p_k = \sin^2((2k+1)\theta/2)$. Evaluate $p_0, \ldots, p_5$, find the optimal k^*, and describe quantitatively what happens at $k = 4, 5$ (over-rotation).
(c) For general $N = 2^n$ with one marked item, give the leading-order optimal iteration count k^* and the resulting query complexity.

Part III — Quantum Fourier transform and phase estimation (Lecture 4)

Problem 7. The Hadamard test. *(10 pts)* Let U be a unitary and $|\psi\rangle$ a state, with the standard Hadamard-test circuit: Hadamard on the ancilla, controlled-U, Hadamard, then measure the ancilla.

(a) Show the ancilla outcome has $p(0) = \frac{1}{2}(1+\Re\langle\psi|U|\psi\rangle)$, and describe the one-gate modification that instead yields $\Im\langle\psi|U|\psi\rangle$.

(b) Take $U = S = \mathrm{diag}(1, \mathrm{i})$ and $|\psi\rangle = |+\rangle$. Compute $\langle\psi|U|\psi\rangle$ directly, then confirm it is consistent with the $p(0)$ (and the imaginary-part variant) of part (a).

Problem 8. QFT and phase estimation. *(10 pts)*

(a) Write the product (tensor-factor) form of $\mathrm{QFT}\,|j\rangle$ on $n = 3$ qubits and count the Hadamard and controlled-phase gates in the circuit.

(b) Run standard QPE with $t = 3$ ancillas on an eigenstate with phase $\varphi = 0.011_2 = \frac{3}{8}$. Write the ancilla state after the controlled powers, identify it as $\mathrm{QFT}\,|k\rangle$ for the correct k, and give the measurement outcome. Why is the readout exact here?

(c) Using the precision/qubit-count relation of §4.3–4.4, determine the number of ancilla qubits t needed to read $n = 6$ bits of φ with success probability at least 0.9. State the formula, then evaluate it.

Part IV — Applications of phase estimation (Lecture 5)

Problem 9. Ground-state energy estimation. *(10 pts)* Let H have eigenpairs $\{(E_j, |E_j\rangle)\}$ and set $U = \mathrm{e}^{-\mathrm{i}Ht}$ for a fixed $t > 0$ chosen so that every $E_j t/(2\pi) \in [0, 1)$.

(a) Show $|E_j\rangle$ is an eigenstate of U with eigenphase $\varphi_j = -E_j t/(2\pi) \bmod 1$, so that a QPE estimate of φ_j yields an estimate of E_j. Give the formula recovering E_j from φ_j.

(b) One prepares an initial guess $|\phi\rangle = \sum_j c_j\,|E_j\rangle$ with ground-state overlap $p_0 = |c_0|^2$ and runs QPE on U. Explain why a single run returns the ground-state phase only with probability p_0, and compute how many independent runs R guarantee seeing it at least once with probability ≥ 0.99 when $p_0 = 0.4$.

(c) Briefly: why does this energy-estimation scheme *not* require a spectral gap, whereas preparing the ground state does?

Problem 10. Amplitude estimation and the quadratic advantage. *(10 pts)* Let $A\,|0^n\rangle = \sqrt{P}\;|\phi_{\mathrm{good}}\rangle + \sqrt{1-P}\;|\phi_{\mathrm{bad}}\rangle$ and write $P = \sin^2(\theta/2)$.

(a) State why phase estimation of the Grover iterate $G = R_{\phi_0} R_{\mathrm{good}}$ (with $R_{\phi_0} = 2\,|\phi_0\rangle\langle\phi_0| - \mathbb{I}$, $R_{\mathrm{good}} = \mathbb{I} - 2\,|\phi_{\mathrm{good}}\rangle\langle\phi_{\mathrm{good}}|$, $|\phi_0\rangle = A\,|0^n\rangle$) estimates θ, and hence P. What are the eigenvalues of G on the invariant plane?

(b) To estimate P to additive error $\epsilon = 10^{-3}$, compare the number of uses of A (and of G) for amplitude estimation, $\mathcal{O}(1/\epsilon)$, against classical Monte Carlo sampling, $\mathcal{O}(1/\epsilon^2)$. Give both counts.

(c) Combine with Problem 7: amplitude estimation applied to the Hadamard-test success probability $p(0) = \frac{1}{2}(1+\Re\langle\psi|U|\psi\rangle)$ estimates $\Re\langle\psi|U|\psi\rangle$. What precision scaling in queries does this achieve, and why is it a quadratic improvement over plain repetition of the Hadamard test?

Problems drawn from Lectures 1 (§1.1–§1.6), 3 (§3.1–§3.3), 4 (§4.1–§4.3), and 5 (§5.1–§5.2) of the Quantum Scientific Computing course notes.